

Trainings ▾

Preparatory Steps

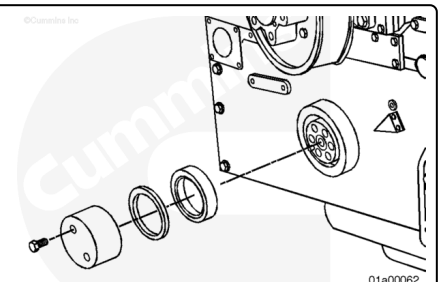
- Disconnect the batteries. Refer to the OEM service manual.
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-040-tr.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html>)
- Remove the vibration damper and fan drive pulley. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/57/57-001-052-tr.html>)



Remove

Remove the front oil seal from the gear cover. Use the front crankshaft seal removal/installation tool kit, Part Number 3163485.

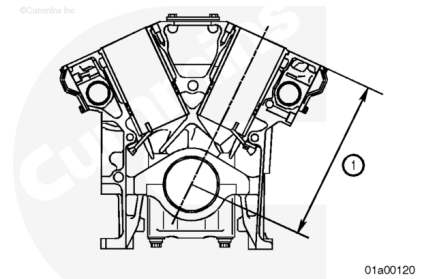
Discard the seal.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



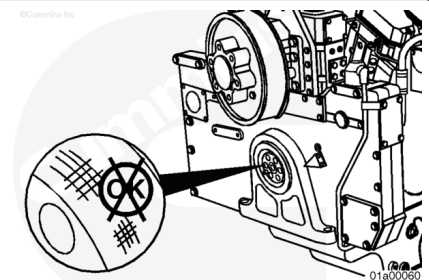
Use a clean cloth and electrical contact cleaner, Part Number 3824510, or equivalent. Clean the crankshaft and seal mounting area.

Check the crankshaft for wear of the oil seal contact surface.

If wear marks are observed, the part **must** be repaired or replaced.

Use the crankshaft reuse guidelines to determine if it is necessary to compensate for crankshaft wear. Refer to Procedure 001-016 in Section 1.

(/qs3/pubsys2/xml/en/procedures/57/57-001-016-tr.html)



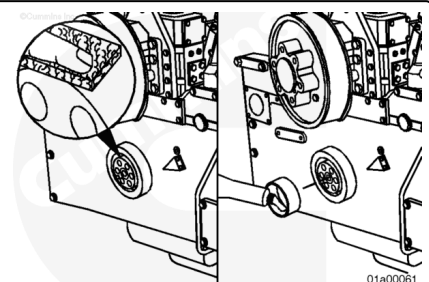
Install

If necessary to compensate for crankshaft wear, install a sleeve, Part Number 3803994, onto the crankshaft.

Clean the first 60 mm [2.36 in] of the crankshaft with Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent.

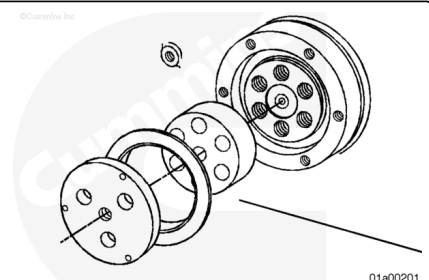
Coat the inside surface of the sleeve with RTV silicone adhesive/sealant, Part Number 4918771, or equivalent.

Use a mallet to tap the sleeve onto the crankshaft. Install the sleeve to a depth of 5 mm [0.20 in] using the front crankshaft seal removal/installation tool kit, Part Number 3163485.



⚠ CAUTION ⚠

Make certain the crankshaft extension is mounted flush and there are no sharp edges protruding. Failure to do so can cause damage to the new crankshaft oil seal.



Note : Lubricants must not be used during the installation process.

The crankshaft extension mounts by using the crankshaft dowel as a guide. Use the front crankshaft installation/removal tool, Part Number 3163485, to install the seal, Part Number 3803994.

Mount the crankshaft extension with the three slotted capscrews. Securely tighten the capscrews.

Note : Make certain the seal is far enough on the crankshaft extension to allow the installation of the installation tool.

Install the installation tool over the crankshaft extension and new oil seal, and onto the end of the crankshaft nose using the existing damper capscrews and appropriate spacers.

Tighten the capscrews until the installation tool makes contact with the oil seal.

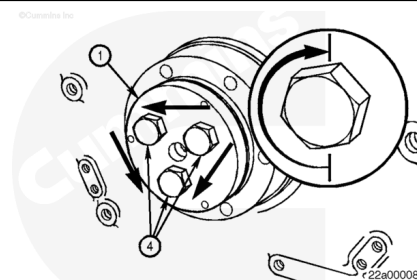
⚠ CAUTION ⚠

The seal must be pressed evenly into the bore area. Failure to do so can result in damage to the seal.

Tighten the capscrews (4) no more than 1/2 of a revolution at a time, alternating capscrews in a **clockwise** pattern until the installation tool (1) makes contact with the crankshaft extension.

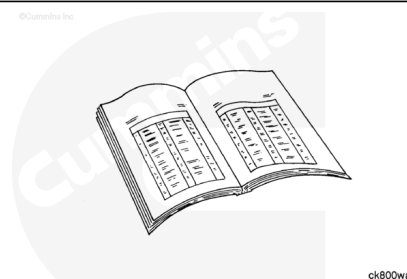
Remove the capscrews (4) and the front crankshaft seal tool (1) from the crankshaft.

Inspect the seal. Verify the yellow dust lip is **not** damaged. The yellow dust seal **must** be positioned 15 mm [0.59 in] from the edge of the gear case cover trunnion.



Finishing Steps

- Install the vibration damper and fan drive pulley. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/57/57-001-052-tr.html>)
- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html>)



- Install the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-040-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Start the engine and check for proper operation and leaks.

Last Modified: 09-Aug-2022
